Q1 2021 (31 Aug Shift 2)

The locus of mid-points of the line segments joining $(-3,-5)$ and the points on the ellipse $\frac{x^2}{4}+\frac{y^2}{9}=1$ is :

(1) $9\mathrm{x}^2+4\mathrm{y}^2+18\mathrm{x}+8\mathrm{y}+145=0$

(2) $36x^2+16y^2+90x+56y+145=0$

(3) $36x^2+16y^2+108x+80y+145=0$

(4) $36\mathrm{x}^2+16\mathrm{y}^2+72\mathrm{x}+32\mathrm{y}+145=0$

Q2 2021 (31 Aug Shift 1)

The line $12x\cos\theta+5y\sin\theta=60$ is tangent to which of the following curves?

(1) $x^2+y^2=169$

(2) $144\mathrm{x}^2+25\mathrm{y}^2=3600$

(3) $25x^2+12y^2=3600$

(4) $x^2+y^2=60$

Q3 2021 (27 Aug Shift 1)

If the minimum area of the triangle formed by a tangent to the ellipse $\frac{x^2}{b^2}+\frac{y^2}{4a^2}=1$ and the co-ordinate axis is kab, then k is equal to .

Q4 2021 (27 Aug Shift 1)

If $x^2+9y^2-4x+3=0, x, y\in\mathbb{R}$, then x and y respectively lie in the intervals:

(1) $\left[-\frac{1}{3},\frac{1}{3}\right]$ and $\left[-\frac{1}{3},\frac{1}{3}\right]$

(2) $\left[-\frac{1}{3},\frac{1}{3}\right]$ and $[1,3]$

(3) $[1,3]$ and $[1,3]$

(4) $[1,3]$ and $\left[-\frac{1}{3},\frac{1}{3}\right]$

Q5 2021 (26 Aug Shift 1)

On the ellipse $\frac{x^2}{8}+\frac{y^2}{4}=1$ let P be a point in the second quadrant such that the tangent at P to the ellipse is perpendicular to the line $x+2y=0$. Let S and S' be the foci of the ellipse and e be its eccentricity. If A is the area of the triangle SPS' then, the value of $\left(5-\mathrm{e}^2\right)$. A is :

(1) 6

(2) 12

(3) 14

(4) 24

Answer Key

Q1 (3) **Q2** (2) **Q3** (2) **Q4** (4)

Q5 (1)

Q1 (3)

General point on $\frac{x^2}{4}+\frac{y^2}{9}=1$ is $A(2\cos\theta, 3\sin\theta)$

given $B(-3,-5)$

midpoint $C\left(\frac{2\cos\theta-3}{2}, \frac{3\sin\theta-5}{2}\right)$

$h=\frac{2\cos\theta-3}{2}; k=\frac{3\sin\theta-5}{2}$

$\Rightarrow\left(\frac{2h+3}{2}\right)^2+\left(\frac{2k+5}{3}\right)^2=1$

$\Rightarrow 36x^2+16y^2+108x+80y+145=0$

Q2 (2)

$12x\cos\theta+5y\sin\theta=60$

$\frac{x\cos\theta}{5}+\frac{y\sin\theta}{12}=1$

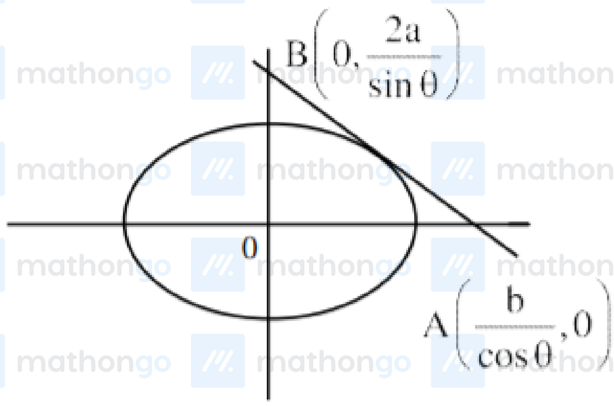
is tangent to $\frac{x^2}{25}+\frac{y^2}{144}=1$

$144x^2+25y^2=3600$

Q3 (2)

Tangent

$\frac{x\cos\theta}{b}+\frac{y\sin\theta}{2a}=1$



So, area $(\Delta OAB)=\frac{1}{2}\times\frac{b}{\cos\theta}\times\frac{2a}{\sin\theta}$

$=\frac{2ab}{\sin 2\theta}\geq 2ab$

$\Rightarrow k=2$

Q4 (4)

$x^2+9y^2-4x+3=0$

$\left(x^2-4x\right)+\left(9y^2\right)+3=0$

$\left(x^2-4x+4\right)+\left(9y^2\right)+3-4=0$

$(x-2)^2+(3y)^2=1$

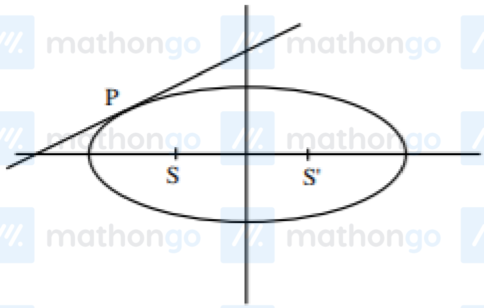
$\frac{(x-2)^2}{(1)^2}+\frac{y^2}{\left(\frac{1}{3}\right)^2}=1$ (equation of an ellipse).

As it is equation of an ellipse, x \& y can vary inside the ellipse.

So, $x-2\in[-1,1]$ and $y\in\left[-\frac{1}{3},\frac{1}{3}\right]$

$x\in[1,3] y\in\left[-\frac{1}{3},\frac{1}{3}\right]$

Q5 (1)



Equation of tangent: $\mathrm{y}=2\mathrm{x}+6$

at P

$\therefore \mathrm{P}(-8/3,2/3)$

$\mathrm{e}=\frac{1}{\sqrt{2}}$

$\mathrm{S}\&\ \mathrm{S}'=(-2,0)\&(2,0)$

Area of Δ SPS $'=\frac{1}{2}\times 4\times\frac{2}{3}$

$A=\frac{4}{3}$

$\therefore\left(5-\mathrm{e}^2\right)\mathrm{A}=\left(5-\frac{1}{2}\right)\frac{4}{3}=6$